# Logarithmic equations

**1** Solve for $y$ in each of the following logarithmic equations:

a $\log_{10} y = 2$

b $\log_{7} y = 5$

c $\log_{\sqrt{49}} (y) = 3$

d $\log_{y} 6 = \frac{1}{2}$

e $\log_{y} \left(\sqrt{6}\right) = 3$

f $\log_{2} y^4 + \log_{2} y = 10$

**2** Solve for $x$ in each of the following logarithmic equations:

a $\log_{16} x = \frac{1}{4}$

b $\log_{25} x = \frac{3}{2}$

c $\log_{4} (5x) = 3$

d $2 \log x = 4$

e $\frac{1}{2} \log_{7} x = \frac{5}{8}$

f $\log_{6} (3x - 9) = 2$

g $\log_{10} (3x + 982) = 3$

h $\log_{9} (7x + 5) = 4$

i $\log_{3} (6 - x) = \frac{1}{2}$

j $6 \log_{4} (2x) - 18 = 0$

k $\log_{7} \left(x^3 - 15\right) = 2$

l $\log_{2} (\log_{2} x) = 0$

m $11 \log_{5} (x - 12) = 33$

n $4 \log_{3} (2x + 1) - 2 = 6$

**3** Solve for $x$ in each of the following logarithmic equations:

a $3 \log x = \log 125$

b $\log_{5} \left(4x^2\right) - \log_{5} x = 3$

c $\log_{10} x + \log_{10} 6 = \log_{10} 48$

d $\log_{10} x - \log_{10} 38 = \log_{10} 37$

e $\log (x + 7) = \log x + \log 3$

f $\log_{10} 12 + \log_{10} x = \log_{10} (x + 6)$

g $\log (x + 9) - \log 2 = \log (7x + 3)$

h $\log (14x - 2) - \log (5x - 2) = \log 3$

i $\log_{8} (5x + 12) = \log_{8} (x + 6) - \log_{8} 3$

j $\log_{2} \left(\sqrt{2x^3}\right) + 1 = 4.5$

k $\log_{7} (2x) + \log_{7} 3 = 3$

l $\log_{3} (8x) - \log_{3} 40 = 2$

m $\log_{4} 45 - \log_{4} (9x) = 2$

n $\log_{5} (x + 12) - \log_{5} (x - 10) = 2$

o $\log_{3} x + \log_{3} (25x) = 8$

**4** Solve for $x$ in each of the following logarithmic equations:

a $\log (x + 5) + \log (x - 2) = \log 8$

b $\log_{6} x = \sqrt{\log_{6} x}$

c $\log x + \log (x + 3) = 1$

d $\log_{10} x + \log_{10} (x - 15) = 2$

e $\log_{4} x + \log_{4} (x - 60) = 4$

f $\log_{4} (x + 3) + \log_{4} (x - 3) = 2$

g $\log_{7} (x + 2) + \log_{7} (x + 6) = \log_{7} 32$